PROJECT REPORT

"SUPERSTORE SALES PERFORMANCE ANALYSIS"

PROJECT OVERVIEW

This project demonstrates the application of **Advanced SQL** to solve real-world business problems. It covers the entire data lifecycle: from **Database Schema Setup** and **Data Cleaning** to **Exploratory Data Analysis (EDA)** and **Actionable Business Insights**.

Key Tech Stack: MySQL | Data Transformation | Business Intelligence

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Introduction

In this project, we have used the dataset of a retail store (Superstore). This dataset contains details of orders, customers, regions, and product categories. By using SQL queries, we have extracted meaningful patterns from the raw data.

Project Objectives

Database Setup: Storing data in a structured format.

Data Cleaning: Fixing inconsistent values and date formats.

Performance Tracking: Calculating overall sales and profit margins.

Strategic Analysis: Identifying loss-making sub-categories and high-value customers.

---------- Step 1 Database Setup ----------

---------- Creating and Selecting a Database ----------

When you create and select a database, no data table appears in the SQL console, rather an Action Output message appears.

```
CREATE DATABASE sales_project;

USE sales_project;
```

---------- Viewing the Table After Data Import ----------

When you run a SELECT * query (with LIMIT 10), you see the table structure and the first 10 records.

```
USE superstore_project;

SELECT * FROM sample_superstore LIMIT 10;
```

Output:

Row_ID	Order_ID	Order_Date	Ship_Date
1	CA-2016-152156	2016-11-08	2016-11-11
2	CA-2016-152156	2016-11-08	2016-11-11
3	CA-2016-138688	2016-06-12	2016-06-16
4	US-2015-108966	2015-10-11	2015-10-18
5	US-2015-108966	2015-10-11	2015-10-18
6	CA-2014-115812	2014-06-09	2014-06-14
7	CA-2014-115812	2014-06-09	2014-06-14
8	CA-2014-115812	2014-06-09	2014-06-14
9	CA-2014-115812	2014-06-09	2014-06-14
10	CA-2014-115812	2014-06-09	2014-06-14

---------- Step 2 Data Cleaning ----------

---------- Turn off Safe Mode so that data can be update --------

In SQL, by default, mass-updating records is restricted. To remove this restriction, we disabled Safe Mode.

```
Set sql_safe_updates = 0;
```

Output:

| `SET sql_safe_updates = 0` | 0 row(s) affected | 0.000 sec |

---------- Converting dates to the correct format (YYYY-MM-DD) --------

In the raw dataset, the dates were in 'Text' format (e.g., '2016-11-08'). We converted them into SQL-readable date format using the STR_TO_DATE function.

```
UPDATE sample_superstore
SET order_date = STR_TO_DATE(order_date, '%Y-%m-%d'),
   ship_date = STR_TO_DATE(ship_date, '%Y-%m-%d');
```

Output:

| `UPDATE sample_superstore` | **Rows matched: 9994, Changed: 9994, Warnings: 0** |

---------- Permanently changing the column's data type to Date ----------

Even after the data is updated, the nature of the column remains 'Text'. ALTER TABLE was used to permanently change the column to DATE type. This will make it easier to run date-based functions (like Month, Year analysis) in the future.

```
    ALTER TABLE sample_superstore
MODIFY COLUMN order_date DATE,
MODIFY COLUMN ship_date DATE;
```

Output:

| `ALTER TABLE sample_superstore` | 0 row(s) affected, Records: 9994 | 0.422 sec |

---------- Step 3 Bussiness Analysis (EDA) ----------

In this step, we have analyzed the overall health and performance of the business using aggregate functions (SUM, COUNT) and the GROUP BY clause.

---------- Total Sales and Profit ----------

This query provides a summary view of the entire dataset, which helps us understand the total revenue and profitability.

```
SELECT
  SUM(sales) AS Total_Revenue,
  SUM(profit) AS Total_Profit,
  (SUM(profit)/SUM(sales))*100 AS Profit_Margin
FROM sample_superstore;
```

Output:

Total_Revenue	Total_Profit	Profit_Margin
2,297,200.86	**286,397.02**	**12.467**

The overall profit margin of the superstore is about 12.47%, which is an indication of a healthy retail business.

---------- Region-wise Analysis ----------

This query tells us which geographical region is the most profitable for business.

```
SELECT region, SUM(sales) as total_sales, SUM(profit) as total_profit

FROM sample_superstore

GROUP BY region

ORDER BY total_profit DESC;
```

Output:

Region	Total_Sales	Total_Profit
West	725,457.82	108,418.45
East	678,781.24	91,522.71
South	391,721.90	46,749.43
Central	501,239.89	39,706.43

The West Region is generating the most profit, while the Central Region has the lowest profit margin despite having good sales.

---------- Profit by Category ----------

This analysis tells us which product lines the business should focus on more.

```
SELECT category, SUM(sales) as total_sales, SUM(profit) as total_profit

FROM sample_superstore

GROUP BY category

ORDER BY total_sales DESC;
```

Output:

Category	Total_Sales	Total_Profit
Technology	836,154.03	145,454.94
Furniture	742,000.12	18,451.27
Office Supplies	719,046.71	122,490.81

The technology category is generating the most revenue and profit. The surprising thing is that despite high sales in furniture, its profit is quite low compared to other categories.

========== Step 4 Problem Solving ----------

In this step, we have targeted the specific areas where the business is incurring losses and identified the customers who are the most valuable for the business.

========== Identifying Loss Making Sub Categories ---------

In business analysis, looking at profit alone is not enough; we also have to see which products are consuming the business's capital. In this query, we have used the HAVING clause to filter only the sub-categories with negative profit (loss).

```
SELECT sub_category, SUM(profit) as net_profit

FROM sample_superstore

GROUP BY sub_category

HAVING net_profit < 0

ORDER BY net_profit ASC;
```

Output:

Sub_Category	Net_Profit	Status
Tables	-$17,725.48	**Critical Loss**
Bookcases	-$3,472.55	**Loss**
Supplies	-$1,189.10	**Marginal Loss**

Business Insight: The tables sub-category is the main cause of losses. The reason could be excessive discounts or high shipping costs. Management should immediately review the pricing strategy of this category.

========== High value Customers ==========

```
SELECT customer_name, SUM(sales) as total_purchase

FROM sample_superstore

GROUP BY customer_name

ORDER BY total_purchase DESC

LIMIT 5;
```

Output:

Customer_Name	Total_Purchase	Rank
Sean Miller	$25,043.05	1
Tamara Chand	$19,052.22	2
Raymond Buch	$15,117.34	3
Tom Ashbrook	$14,595.62	4
Adrian Barton	$14,473.57	5

Business Insight: These 5 customers are 'High-Value' assets for the Superstore. Monitoring the frequency of their orders and their satisfaction level is essential for business growth.

========== The "Grand Final" Query ==========

The purpose of this query is to see which category is performing best in which region and where we are making the most profit or loss.

```
SELECT
    category,
    region,
    COUNT(order_id) as total_orders,
    ROUND(SUM(sales), 2) as total_revenue,
    ROUND(SUM(profit), 2) as net_profit
FROM sample_superstore
GROUP BY category, region
ORDER BY net_profit DESC;
```

Output:

Category	Region	Total_Orders	Total_Revenue
Technology	**East**	453	264,973.98
Technology	**West**	480	251,991.83
Office Supplies	**West**	1,897	220,853.25
Technology	**South**	293	148,771.91
Office Supplies	**East**	1,712	205,516.05
Furniture	**West**	507	252,612.74

Technology	**Central**	421	170,416.31
Office Supplies	**Central**	1,422	167,026.41
Furniture	**East**	601	208,291.21
Furniture	**South**	332	117,298.68
Furniture	**Central**	481	163,797.16

Conclusion:

Through SQL analysis, we transformed raw data into actionable insights. We found that the business is overall profitable (12.47% margin), but immediate action is needed to bring the Furniture category out of loss in the Central region. Additionally, by focusing on Technology and Office Supplies in the West and East regions, revenue can be further increased.